The principal components of EOG's rollforward of valuation allowances for deferred income tax assets for the years indicated below were as follows (in thousands):

	2018	2017	2016
Beginning Balance	$ 466,421	$ 383,221	$ 506,127
Increase [(1)]	23,062	67,333	37,221
Decrease [(2)]	(26,219)	(13,687)	(12,667)
Other [(3)]	(296,122)	29,554	(147,460)
Ending Balance	**$ 167,142**	**$ 466,421**	**$ 383,221**

(1) Increase in valuation allowance related to the generation of tax NOLs and other deferred tax assets.
(2) Decrease in valuation allowance associated with adjustments to certain deferred tax assets and their related allowance.
(3) Represents dispositions, revisions and/or foreign exchange rate variances and the effect of statutory income tax rate changes. The United Kingdom operations were sold in the fourth quarter of 2018. The Argentina operations were sold in the third quarter of 2016.

As of December 31, 2018, EOG had state income tax NOLs being carried forward of approximately $1.8 billion, which, if unused, expire between 2019 and 2037. EOG also has Canadian NOLs of $183 million which can be carried forward 20 years. As described above, these NOLs as well as other less significant future tax benefits, have been evaluated for the likelihood of utilization, and valuation allowances have been established for the portion of these deferred income tax assets that do not meet the “more likely than not” threshold.

The balance of unrecognized tax benefits at December 31, 2018, was $29 million, resulting from the tax treatment of its research and experimental expenditures related to certain innovations in its horizontal drilling and completion projects, of which $12 million may potentially have an earnings impact. EOG records interest and penalties related to unrecognized tax benefits to its income tax provision. Currently $2 million of interest has been recognized in the Consolidated Statements of Income (Loss) and Comprehensive Income (Loss). EOG does not anticipate that the amount of the unrecognized tax benefits will change materially during the next twelve months. EOG and its subsidiaries file income tax returns and are subject to tax audits in the U.S. and various state, local and foreign jurisdictions. EOG's earliest open tax years in its principal jurisdictions are as follows: U.S. federal (2016), Canada (2014), Trinidad (2013) and China (2008).

EOG's foreign subsidiaries' undistributed earnings are not considered to be permanently reinvested outside of the U.S. Accordingly, EOG may be required to accrue certain U.S. federal, state, and foreign deferred income taxes on these undistributed earnings as well as on any other outside basis differences related to its investments in these subsidiaries. As of December 31, 2018, EOG has cumulatively recorded $23 million of deferred foreign income taxes for withholdings on its undistributed foreign earnings. Additionally, for tax years beginning in 2018 and later, EOG's foreign earnings may be subject to the U.S. federal "global intangible low-taxed income" (GILTI) inclusion. EOG records any GILTI tax as a period expense.

7. Employee Benefit Plans

Stock-Based Compensation

During 2018, EOG maintained various stock-based compensation plans as discussed below. EOG recognizes compensation expense on grants of stock options, SARs, restricted stock and restricted stock units, performance units and grants made under the EOG Resources, Inc. Employee Stock Purchase Plan (ESPP). Stock-based compensation expense is calculated based upon the grant date estimated fair value of the awards, net of forfeitures, based upon EOG's historical employee turnover rate. Compensation expense is amortized over the shorter of the vesting period or the period from date of grant until the date the employee becomes eligible to retire without company approval.